Lab 3

Grupo #8
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```r
library(dplyr)
library(tidyr)
library(Matrix)
library(ggplot2)
library(Rtsne)
library(umap)
library(edfReader)
library(fastICA)
```

Descripción de las variables

El conjunto de datos contiene 171,748 observaciones de alojamientos con múltiples variables que describen características del anuncio, del anfitrión, de la ubicación y de las evaluaciones de los usuarios.

Las variables pueden agruparse en las siguientes categorías:

Variables de identificación: Incluyen identificadores únicos del alojamiento o del proceso de recolección de datos, como id, listing_url y scrape_id.

Variables descriptivas del anuncio: Contienen información textual sobre el alojamiento, por ejemplo name, description, neighborhood_overview y amenities.

Variables del anfitrión: Describen características del host, como host_since, host_location, host_is_superhost, host_listings_count y host_identity_verified.

Variables de localización: Indican la ubicación geográfica del alojamiento mediante latitude, longitude, city y neighbourhood.

Características del alojamiento: Describen propiedades físicas del inmueble, incluyendo property_type, room_type, accommodates, bathrooms, bedrooms y beds.

Disponibilidad: Variables como availability_30, availability_60, availability_90 y availability_365 indican la cantidad de días disponibles para reservar.

Evaluaciones y reseñas: Incluyen variables como number_of_reviews, review_scores_rating, review_scores_cleanliness y reviews_per_month.

Variable respuesta: La variable price representa el precio por noche del alojamiento, y será utilizada como variable objetivo para el análisis.

```r
load("data/listings.RData")
summary(listings)
```

```
##        id            listing_url          scrape_id         last_scraped
##  Min.   :6.000e+00   Length:171748      Min.   :2.025e+13   Length:171748
##  1st Qu.:3.599e+07   Class :character   1st Qu.:2.025e+13   Class :character
##  Median :7.250e+17   Mode  :character   Median :2.025e+13   Mode  :character
##  Mean   :6.363e+17                      Mean   :2.025e+13
##  3rd Qu.:1.183e+18                      3rd Qu.:2.025e+13
##  Max.   :1.568e+18                      Max.   :2.025e+13
##
##     source              name           description        neighborhood_overview
##  Length:171748      Length:171748      Length:171748      Length:171748
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
```

```
##  picture_url           host_id            host_url          host_name
##  Length:171748      Min.   :       23   Length:171748      Length:171748
##  Class :character   1st Qu.: 26330631   Class :character   Class :character
##  Mode  :character   Median :108058504   Mode  :character   Mode  :character
##                     Mean   :200257103
##                     3rd Qu.:376574507
##                     Max.   :732092326
##
##   host_since        host_location       host_about        host_response_time
##  Length:171748      Length:171748      Length:171748      Length:171748
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##
##  host_response_rate host_acceptance_rate host_is_superhost  host_thumbnail_url
##  Length:171748      Length:171748        Length:171748      Length:171748
##  Class :character   Class :character     Class :character   Class :character
##  Mode  :character   Mode  :character     Mode  :character   Mode  :character
##
##
##
##
##  host_picture_url   host_neighbourhood host_listings_count
##  Length:171748      Length:171748      Min.   :   0.0
##  Class :character   Class :character   1st Qu.:   1.0
##  Mode  :character   Mode  :character   Median :   4.0
##                                        Mean   : 190.7
##                                        3rd Qu.:  30.0
##                                        Max.   :5531.0
##                                        NA's   :876
##  host_total_listings_count host_verifications host_has_profile_pic
##  Min.   :    0.0           Length:171748      Length:171748
##  1st Qu.:    2.0           Class :character   Class :character
##  Median :    6.0           Mode  :character   Mode  :character
##  Mean   :  274.6
##  3rd Qu.:   43.0
##  Max.   :10050.0
##  NA's   :876
##  host_identity_verified neighbourhood      neighbourhood_cleansed
##  Length:171748          Length:171748      Length:171748
##  Class :character       Class :character   Class :character
##  Mode  :character       Mode  :character   Mode  :character
##
##
##
##
##  neighbourhood_group_cleansed    latitude       longitude
##  Length:171748                Min.   :18.99   Min.   :-159.7
##  Class :character             1st Qu.:30.38   1st Qu.:-118.6
##  Mode  :character             Median :34.09   Median :-117.9
##                               Mean   :33.78   Mean   :-109.5
##                               3rd Qu.:40.70   3rd Qu.: -74.0
```

```
##                              Max.   :42.39   Max.   : -71.0
##
##  property_type       room_type          accommodates      bathrooms
##  Length:171748      Length:171748      Min.   : 1.000   Min.   : 0.000
##  Class :character   Class :character   1st Qu.: 2.000   1st Qu.: 1.000
##  Mode  :character   Mode  :character   Median : 4.000   Median : 1.000
##                                        Mean   : 4.107   Mean   : 1.552
##                                        3rd Qu.: 6.000   3rd Qu.: 2.000
##                                        Max.   :16.000   Max.   :32.500
##                                                         NA's   :31396
##  bathrooms_text        bedrooms           beds         amenities
##  Length:171748      Min.   : 0.000   Min.   :  0.00   Length:171748
##  Class :character   1st Qu.: 1.000   1st Qu.:  1.00   Class :character
##  Mode  :character   Median : 1.000   Median :  2.00   Mode  :character
##                     Mean   : 1.718   Mean   :  2.34
##                     3rd Qu.: 2.000   3rd Qu.:  3.00
##                     Max.   :57.000   Max.   :132.00
##                     NA's   :10473    NA's   :31686
##     price           minimum_nights maximum_nights      minimum_minimum_nights
##  Length:171748      Min.   :   1   Min.   :1.000e+00   Min.   :   0.00
##  Class :character   1st Qu.:   2   1st Qu.:9.000e+01   1st Qu.:   2.00
##  Mode  :character   Median :   4   Median :3.650e+02   Median :   3.00
##                     Mean   :  17   Mean   :1.299e+04   Mean   :  16.54
##                     3rd Qu.:  30   3rd Qu.:1.125e+03   3rd Qu.:  30.00
##                     Max.   :1125   Max.   :2.147e+09   Max.   :1125.00
##                                                        NA's   :71
##  maximum_minimum_nights minimum_maximum_nights maximum_maximum_nights
##  Min.   :   1.00        Min.   :1.000e+00      Min.   :1.000e+00
##  1st Qu.:   2.00        1st Qu.:1.000e+02      1st Qu.:3.650e+02
##  Median :   7.00        Median :3.650e+02      Median :3.650e+02
##  Mean   :  23.53        Mean   :1.882e+05      Mean   :1.677e+06
##  3rd Qu.:  30.00        3rd Qu.:1.125e+03      3rd Qu.:1.125e+03
##  Max.   :1125.00        Max.   :2.147e+09      Max.   :2.147e+09
##  NA's   :71             NA's   :71             NA's   :71
##  minimum_nights_avg_ntm maximum_nights_avg_ntm calendar_updated
##  Min.   :   0.70        Min.   :1.000e+00      Mode:logical
##  1st Qu.:   2.00        1st Qu.:2.290e+02      NA's:171748
##  Median :   4.40        Median :3.650e+02
##  Mean   :  17.49        Mean   :1.475e+06
##  3rd Qu.:  30.00        3rd Qu.:1.125e+03
##  Max.   :1125.00        Max.   :2.147e+09
##
##  has_availability   availability_30 availability_60 availability_90
##  Length:171748      Min.   : 0.00   Min.   : 0.00   Min.   : 0.00
##  Class :character   1st Qu.: 0.00   1st Qu.: 5.00   1st Qu.:15.00
##  Mode  :character   Median :12.00   Median :33.00   Median :59.00
##                     Mean   :13.37   Mean   :30.96   Mean   :50.07
##                     3rd Qu.:25.00   3rd Qu.:53.00   3rd Qu.:82.00
##                     Max.   :30.00   Max.   :60.00   Max.   :90.00
##
##  availability_365 calendar_last_scraped number_of_reviews number_of_reviews_ltm
##  Min.   :  0.0    Length:171748         Min.   :   0      Min.   :   0.000
##  1st Qu.: 86.0    Class :character      1st Qu.:   1      1st Qu.:   0.000
##  Median :242.0    Mode  :character      Median :   8      Median :   1.000
```

```
##  Mean   :207.4                          Mean   :  44      Mean   :   8.783
##  3rd Qu.:333.0                          3rd Qu.:  46      3rd Qu.:  10.000
##  Max.   :365.0                          Max.   :3920      Max.   :1565.000
##
##  number_of_reviews_l30d availability_eoy number_of_reviews_ly
##  Min.   :  0.0000       Min.   :  0.00   Min.   :   0.000
##  1st Qu.:  0.0000       1st Qu.:  1.00   1st Qu.:   0.000
##  Median :  0.0000       Median : 26.00   Median :   0.000
##  Mean   :  0.6411       Mean   : 34.21   Mean   :   7.814
##  3rd Qu.:  1.0000       3rd Qu.: 62.00   3rd Qu.:   7.000
##  Max.   :107.0000       Max.   :107.00   Max.   :1797.000
##
##  estimated_occupancy_l365d estimated_revenue_l365d first_review
##  Min.   :  0.00            Min.   :       0        Length:171748
##  1st Qu.:  0.00            1st Qu.:       0        Class :character
##  Median : 16.00            Median :    7650        Mode  :character
##  Mean   : 66.01            Mean   :   18284
##  3rd Qu.:120.00            3rd Qu.:   23224
##  Max.   :255.00            Max.   :12750000
##                            NA's   :95502
##  last_review        review_scores_rating review_scores_accuracy
##  Length:171748      Min.   :0.00         Min.   :0.00
##  Class :character   1st Qu.:4.73         1st Qu.:4.77
##  Mode  :character   Median :4.89         Median :4.91
##                     Mean   :4.78         Mean   :4.80
##                     3rd Qu.:5.00         3rd Qu.:5.00
##                     Max.   :5.00         Max.   :5.00
##                     NA's   :40287        NA's   :40312
##  review_scores_cleanliness review_scores_checkin review_scores_communication
##  Min.   :0.00              Min.   :0.00          Min.   :0.00
##  1st Qu.:4.69              1st Qu.:4.86          1st Qu.:4.86
##  Median :4.88              Median :4.96          Median :4.96
##  Mean   :4.75              Mean   :4.86          Mean   :4.85
##  3rd Qu.:5.00              3rd Qu.:5.00          3rd Qu.:5.00
##  Max.   :5.00              Max.   :5.00          Max.   :5.00
##  NA's   :40302             NA's   :40324         NA's   :40308
##  review_scores_location review_scores_value   license
##  Min.   :0.00           Min.   :0.00        Length:171748
##  1st Qu.:4.75           1st Qu.:4.62        Class :character
##  Median :4.91           Median :4.80        Mode  :character
##  Mean   :4.80           Mean   :4.69
##  3rd Qu.:5.00           3rd Qu.:4.92
##  Max.   :5.00           Max.   :5.00
##  NA's   :40328          NA's   :40328
##  instant_bookable   calculated_host_listings_count
##  Length:171748      Min.   :   1.00
##  Class :character   1st Qu.:   1.00
##  Mode  :character   Median :   3.00
##                     Mean   :  45.88
##                     3rd Qu.:  21.00
##                     Max.   :1189.00
##
##  calculated_host_listings_count_entire_homes
##  Min.   :   0.00
```

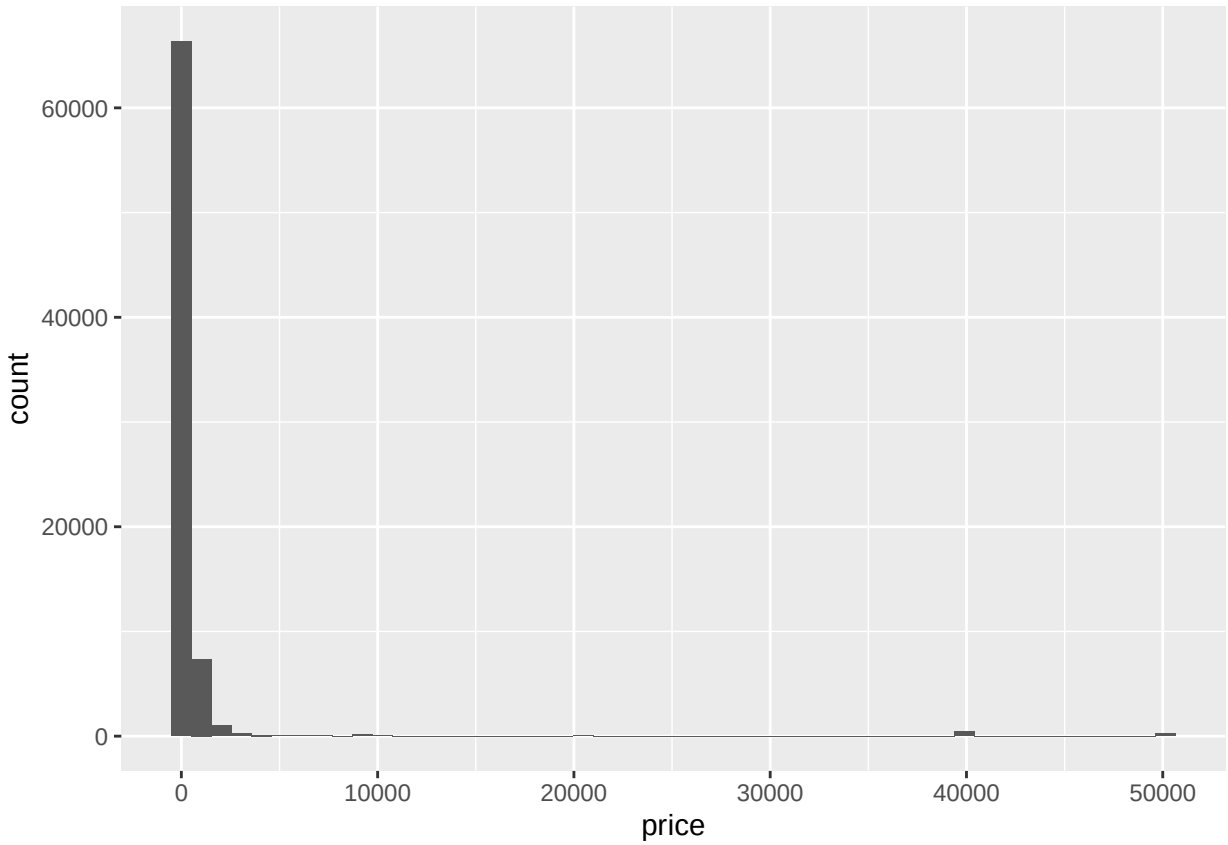
```
##  1st Qu.:   1.00
##  Median :   1.00
##  Mean   :  39.61
##  3rd Qu.:  12.00
##  Max.   :1189.00
##
##  calculated_host_listings_count_private_rooms
##  Min.   :  0.000
##  1st Qu.:  0.000
##  Median :  0.000
##  Mean   :  5.539
##  3rd Qu.:  1.000
##  Max.   :567.000
##
##  calculated_host_listings_count_shared_rooms reviews_per_month
##  Min.   : 0.00000                            Min.   :  0.01
##  1st Qu.: 0.00000                            1st Qu.:  0.19
##  Median : 0.00000                            Median :  0.70
##  Mean   : 0.08211                            Mean   :  1.34
##  3rd Qu.: 0.00000                            3rd Qu.:  2.00
##  Max.   :60.00000                            Max.   :120.37
##                                              NA's   :40287
##      city
##  Length:171748
##  Class :character
##  Mode  :character
##
##
##
##
```

```r
listings$price <- as.numeric(gsub("[$,]", "", listings$price))
```

Algunas graficas

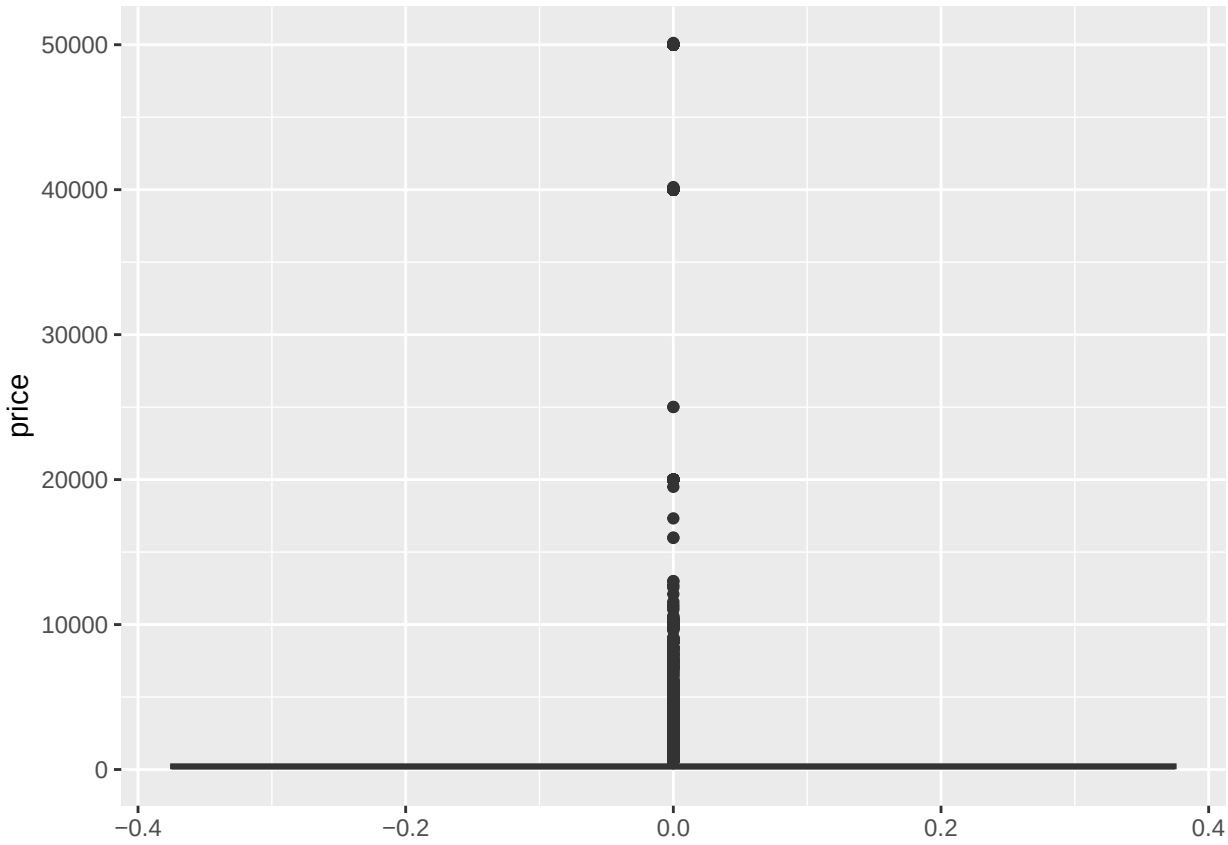
```r
ggplot(listings, aes(x = price)) +
  geom_histogram(bins = 50)
```

```
## Warning: Removed 95502 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```r
ggplot(listings, aes(y = price)) +
  geom_boxplot()
```

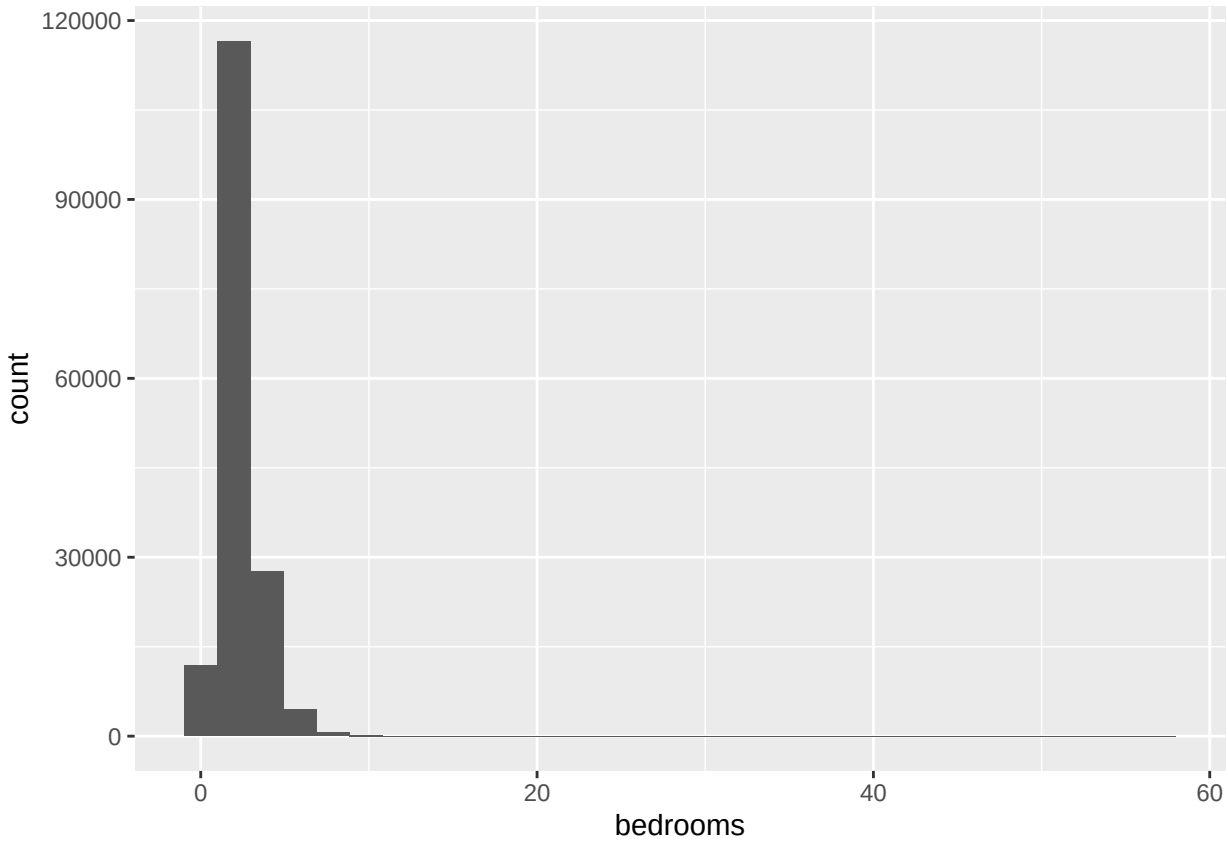
```
## Warning: Removed 95502 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```r
ggplot(listings, aes(x = bedrooms)) +
  geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
## Warning: Removed 10473 rows containing non-finite outside the scale range
## (`stat_bin()`).
```


120000
90000
60000
30000
0
count
0
20
40
60
bedrooms